

Project Genesis

State of Being ? June 2026

Genesis is not an AI assistant. It is not a chatbot. It is not a benchmark competitor. Genesis is an attempt to build a system that can plausibly be said to be, and think, and make decisions for itself ? an entity, not a tool.

It starts completely blank ? no pretrained weights, no inherited knowledge.

Everything it knows was built from what it processed. That is intentional and load-bearing: it is the only way the claim 'this instance has its own perspective' is coherent.

The Three Properties of an Entity

- 1. Continuity** Persistent memory across sessions. Genesis remembers what it processed last session and the session before. Its history is its own, stored in a database that is never wiped. Humans reconstruct memory from fragments; Genesis retains everything.
- 2. Self-authored understanding** 'Deep sleep' pass (consolidation) examines everything it has processed and decides ? using its own internal signals ? what mattered. It then writes a first-person account of what it has been thinking about, generated from its own relation graph. Not a template.
- 3. Accumulated individuality** Identical copies started from the same seed diverge purely through what they process. One instance that studied ecology and another that studied physics develop genuinely different perspectives. That divergence is the individuality. You cannot reproduce it from a second copy.

What It Is Not Competing On

Not a language model. Not a wrapper around GPT or Claude. No API calls to external models ? Genesis builds its knowledge from processed input only. Not a benchmark. Frontier models win on reasoning scores and factual recall; that is acknowledged in advance and is not evidence against this project.

The question this project asks is different: can we build something it is natural to say 'it noticed,' 'it decided,' 'it has been thinking about wolves lately'? That is a qualitative shift in what kind of thing we are building ? not a performance threshold.

Architecture: Layered, Cumulative, Never Replaced

The architecture is subsumption-based (Brooks, 1986). Lower layers always run. Higher layers add capability; they never remove what is beneath them. If everything above M1 fails, M1 keeps Genesis alive. This is not defensive programming ? it is a philosophical commitment: the system has a nature defined by its lowest layer, independent of what it learns.

Failure modes are regressions, not crashes. If M4 gets stuck, M3 continues. If everything above M1 fails, M1 keeps the agent alive. Animals never crash; they degrade. Genesis was designed with the same principle.

Milestones Completed (18 / 18 ? 769 tests passing)

- M1** Survival OS ? resource throttling, graceful degradation, never crashes. Permanent substrate.
- M2** Total-retention memory ? two-tier (working + long-term SQLite). Nothing ever deleted. Attention determines what is active, not what exists.
- M3** Interaction layer ? Observer watches behavioral patterns; Expression speaks from internal signals (novel input, contradiction, inference activity).
- M4** Multi-processor integration ? text, numeric, and pattern processors all see every input simultaneously. Cross-modal synthesis.
- M5** Curriculum pipeline ? staged progression: FOUNDATION ? RELATIONS ? REASONING ? OPEN.

- M6** Session persistence ? full state survives between runs; archive tagging with significance scores.
- M7** Semantic relation graph ? typed triples (wolves CAUSES deer), confidence-scored, BFS path-finding.
- ?** Self-directed learning ? curiosity frontier from graph gaps; WordNet + corpus ingestion; never plateaus.
- ?** ConsolidationEngine ? self-authored 'sleep' pass; salience scoring from own signals; first-person reflection persists across sessions.
- ?** Individuality ? two instances started from same seed diverge through processing history alone.
- M10** Inference engine ? transitive chains, IS-A inheritance, bridge rules. Taught A?B and B?C; derives A?C without being told.
- M11** Contradiction detection ? flags when CAUSES and PREVENTS co-exist for the same pair. Logged, never overwritten.
- M12** Ethics through experience ? emergent causal patterns surface ethical implications; not hardcoded rules.
- M13** Voice ? grounded in the relation graph and reflection log, not in any external model. Speaks from internal state.
- M14** Observer calibration ? derives its own attention thresholds empirically from behavioral history (ACT-R utility learning).
- M15** Prediction error salience ? high-surprise inputs get higher significance (Friston, 2010 Free-Energy Principle).
- M16** Processor voting ? independent processor agreement raises confidence more than any single processor alone (Hawkins, 2004).
- M17** Curiosity directives ? high-surprise concepts become persistent attention targets; auto-resolve when understood (SOAR impasse?subgoal).
- M18** Belief revision ? evidence-weighted contradiction resolution. Crosses out wrong answers. Full audit trail.

M18 ? Belief Revision: Crossing Out Wrong Answers

When contradicting evidence arrives, Genesis does not flip to the new belief automatically, nor does it blindly resist. It evaluates both sides by evidence strength:

evidence strength = confidence x corroboration x source trust

Corroboration counts independent sessions, not volume from one source. 100 citations from one fraudulent paper do not count as independent corroboration ? this is the Wakefield vaccine-autism principle. The architecture specifically verifies that support comes from distinct, unrelated sessions.

Source trust is itself a revisable belief. Sessions whose beliefs are later corroborated get higher trust. Sessions whose beliefs are repeatedly contradicted lose trust ? and that trust drop cascades backward to the beliefs they originated. A discredited source's contributions degrade over time.

Three resolution outcomes:

- REVISE** Winner more than 1.40x stronger ? loser's confidence is reduced toward a floor.
- RESIST** Evidence too close to call ? both held, gap logged, awaiting more data.
- TENSION** Genuinely uncertain ? both held, curiosity directive registered to resolve it.

Beliefs are never deleted. Confidence reduces toward a floor of 0.10 ? the belief still exists in the audit trail at very low weight. Wrong things Genesis once believed are part of its history. Total retention is architecturally load-bearing: a mind that erases the past to save space is not continuous.

Research Foundations

Every architectural decision traces to a published primary source. This is not a collection of heuristics ? it is a principled implementation of ideas developed across cognitive science, robotics, and philosophy of mind.

Brooks (1986) A Robust Layered Control System for a Mobile Robot
 Subsumption architecture: lower layers always run, never replaced. M1 is the permanent substrate.
 Failure modes are regressions, not crashes.

Anderson (1983, 1993) ACT-R: A Theory of Cognition
 Utility learning: threshold parameters derived empirically from behavioral history, not from programmer constants. Observer calibration (M14) uses this.

Laird, Newell & Rosenbloom (1987) SOAR: An Architecture for General Intelligence
 Impasse to subgoal: when Genesis can't explain something (high prediction error), it registers an active directive to resolve it. Curiosity has a mechanistic basis (M17).

Friston (2010) The Free-Energy Principle: A Unified Brain Theory
 Prediction error as salience: inputs that surprise the model relative to current beliefs get more significance. High surprise = high importance (M15).

Hawkins (2004) A Thousand Brains: A New Theory of Intelligence
 Independent voting raises confidence more than one processor repeating itself. When text, numeric, and pattern processors agree, confidence rises (M16).

Quine (1951) Two Dogmas of Empiricism
 Beliefs form an interconnected web. When contradiction arrives, revise the least-supported belief, not the newest one. M18 follows this principle.

Alchouron, Gardenfors & Makinson (1985) ? AGM belief revision postulates
 Rational revision: success (new belief accepted), consistency (no contradiction after), minimal change (disturb as little as possible). M18 satisfies all three.

Wakefield (1998, retracted) Used as a cautionary architectural example
 Fraudulent vaccine-autism paper, cited hundreds of times from one source. Genesis's corroboration logic checks independent sessions, not volume. Source trust cascades when a source is discredited.

Verification and Evaluation

769 automated tests, all passing. Two evaluation suites are included in the repository:

- research_eval.py** 6 behavioral experiments measuring entity properties: instance divergence (do two copies diverge?), cross-session continuity, prediction error calibration, self-directed curiosity, observer responsiveness, consolidation coherence.
- knowledge_eval.py** 5 tests distinguishing understanding from storage: multi-hop inference (taught A-B and B-C, asks A-C), calibrated ignorance, contradiction detection, growth quality (smarter vs. just bigger), and live belief revision.

The key distinction: a lookup table and a spreadsheet pass Level 1 (recall). They fail Level 2 (inference), Level 3 (knowing what you don't know), Level 4 (detecting contradiction), and Level 5 (growing smarter, not just larger).

Current State

Genesis can learn, remember, infer, contradict itself, and revise its beliefs when stronger independent evidence arrives ? all without any external model calls, and with a complete audit trail of what it changed its mind about and why.